Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-7)

INSTRUMENTAL METHOD OF ANALYSIS-THEORY

Subject Code : BP-701T

M.Code : 78387

Date of Examination : 05-12-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Write briefly :**

   a. Define bathochromic and hypsochromic shift.

   b. What is principle of chromatography?

   c. Give two properties of ion exchange resins.

   d. Define R_f value?

   e. Define fingerprint region.

   f. What is the principle of nepheloturbidometry?

   g. Write two application of flame photometry.

   h. Define singlet and triplet state.

   i. Define partition coefficient.

   j. Define electrophoresis.

SECTION-B

2. Describe the principle, instrumentation and applications of Atomic Absorption Spectroscopy.

3. Explain the theory, instrumentation, and applications of Gas Chromatography.

4. Describe the sources of radiation, wavelength selectors, and detectors used in IR spectroscopy.

SECTION-C

5. Explain the concept of electronic transitions and chromophores in UV-visible spectroscopy.

6. Discuss the factors affecting fluorescence and quenching in fluorimetry.

7. What are the sources of radiation, wavelength selectors and detectors used in Flame Photometry?

8. Explain the methodology and applications of Thin Layer Chromatography.

9. What is the theory behind ion exchange chromatography? Discuss the properties of ion exchange resins.

10. Describe the methodology and applications of Gel chromatography.

11. Discuss the factors affecting electrophoretic mobility in Electrophoresis.

12. Write a note on various vibrations in IR spectroscopy.

13. Explain the concept of Beer and Lambert's law and its deviations in UV-visible spectroscopy.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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